

Intel launches 10th generation of Intel Core H-series chipsets

Intel has launched its latest generation of the 5G-enabled chipset. It has introduced a new 10th generation of Intel Core H-series. The company said that the Intel Core H-series gives faster performance with up to 5.3GHz Turbo with eight cores and sixteen threads. It also said that the chipset provides lower latency and provides the best gaming experience on a laptop.

Intel stated that the Core i9-10980HK chipset gives up to 54 per cent more frames per second in gaming and 44 per cent overall performance as compared to its predecessor. It

also delivers two times faster 4K video rendering and export compared to the previous generation of processors.

The 10th Gen Intel Core i7-10750H processor comes with up to 5.0GHz Turbo. The company claimed that it will give up to 44 per cent more frames per second in gaming, up to 33 per cent better overall performance and up to 70 per cent faster 4K video exporting. Intel also introduced the new 10th Gen Intel Core i7-10875H. It features up to 5.1GHz Turbo, eight cores and sixteen threads.

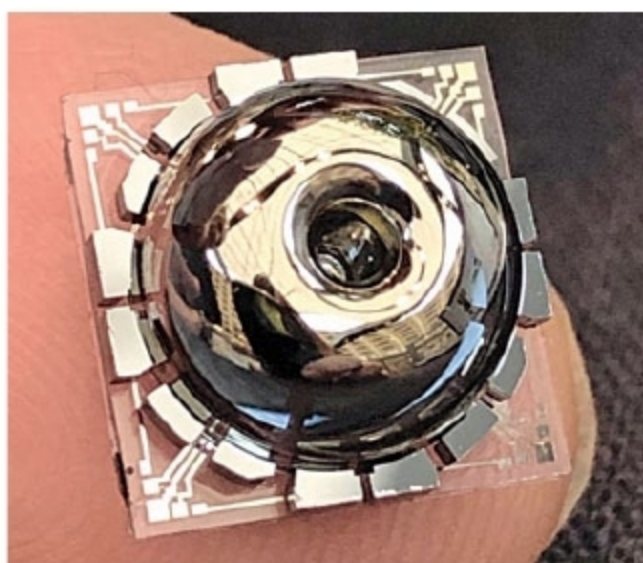
Small, precise and affordable gyroscope for navigating with GPS

A small, inexpensive and highly accurate gyroscope, developed at the University of Michigan, could help drones and autonomous cars stay on track without a GPS signal.

"Our gyroscope is 10,000 times more accurate but only ten times more expensive than gyroscopes used in your typical cell phones. This gyroscope is 1,000 times less expensive than much larger gyroscopes with similar performance," said Khalil Najafi, the Schlumberger Professor of Engineering at U-M and a professor of electrical engineering and computer science.

Most smartphones contain gyroscopes to detect the orientation of the screen and help figure out which way we're facing, but their accuracy is poor. They're the reason why phones often incorrectly indicate which direction a user is facing during navigation.

It doesn't matter much to a human on the street or behind the wheel, but a driverless car could get lost quickly with a loss of GPS signal. Inside their backup navigation systems,



The new resonator and electrodes on a finger for scale. The resonator is almost perfectly symmetrical, made of nearly-pure glass (Credit: Najafi Group)

autonomous vehicles currently use high-performance gyroscopes that are larger and much more expensive.

"High-performance gyroscopes are a bottleneck, and they have been for a long time. This gyroscope can remove this bottleneck by enabling the use of high-precision and low-cost inertial navigation in most autonomous vehicles," said Jae Yoong Cho, an assistant research scientist in electrical engineering and computer science.

The key to making this affordable, small gyroscope is a nearly symmetrical mechanical resonator. It looks like a Bundt pan crossed with a wine glass, made one centimetre wide. As with

wine glasses, the duration of the ringing tone produced when the glass is struck depends on the quality of the glass—but instead of being an aesthetic feature, the ring is crucial to the gyroscope's function. The complete device uses electrodes placed around the glass resonator to push and pull on the glass, making it ring and keep it going.

Smart and customised light solutions based on open source and AI

The research project OpenLicht was launched in September 2016 and stands for the design of smart, customised light solutions based on open source and new materials. OpenLicht was funded by the German Ministry of Education and Research (BMBF), with the goal of enabling new forms of collaboration between science, business, maker and startup community. Infineon Technologies AG is supervising the project in close cooperation with Bernitz Electronics GmbH, Deggendorf Institute of Technology and the Technical University of Dresden.

The results are now being made public and include the prototype of a smart lighting system based on artificial intelligence (AI). It automatically adjusts the light in the room to

the user's position and activity, such as reading or watching TV, learns the person's preferences and can even respond to a certain degree to circumstances it has not learned previously.

The solution developed in the project is based on open-source approaches like openHAB, a smart home system, and machine learning libraries. The use of freely available development environments, software frameworks and low-cost hardware solutions enable integration of a wide range of different sensor data and further development of existing results by the community. Intelligent light design is entering the smart home, so the vision will accompany people through their day-to-day life with fully automated solutions.